

 Marwadi University Marwadi Chandarana Group	NAAC A+	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)		Aim: Simulate discrete time sequences.	
Experiment No: 1	Date:17-07-2026	Enrollment No:92400133030	

Aim: Simulate discrete time sequences.

Github:

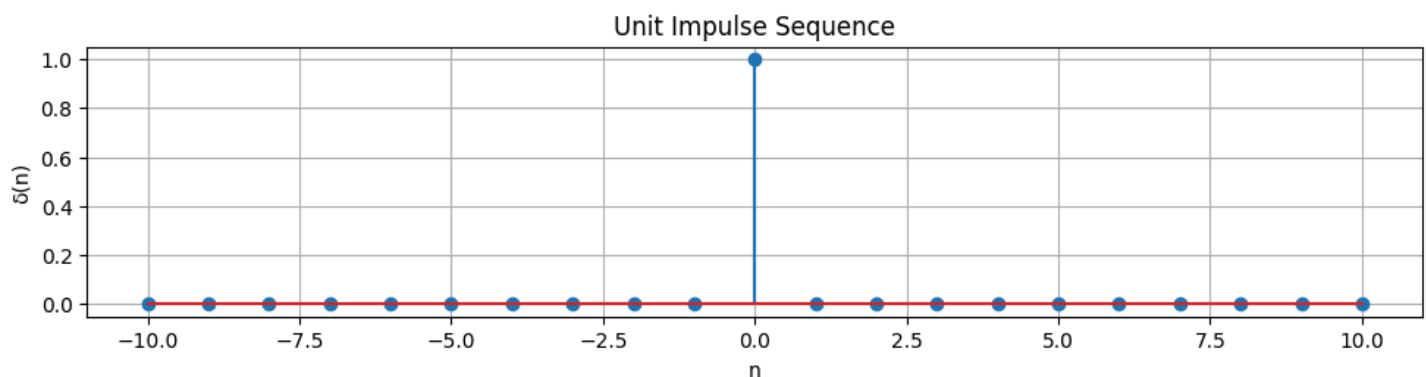
Code 1:

```
import numpy as np
import matplotlib.pyplot as plt
n = np.arange(-10, 11)
impulse = np.where(n == 0, 1, 0)
step = np.where(n >= 0, 1, 0)
ramp = np.where(n >= 0, n, 0)

a = 0.8
exponential = np.where(n >= 0, a**n, 0)
f = 0.1 # Frequency
sinusoidal = np.sin(2 * np.pi * f * n)
plt.figure(figsize=(10, 12))

# Unit Impulse
plt.subplot(5, 1, 1)
plt.stem(n, impulse)
plt.title("Unit Impulse Sequence")
plt.xlabel("n")
plt.ylabel(" $\delta(n)$ ")
plt.grid(True)
plt.tight_layout()
plt.show()
```

Output:



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Code 2:

```

import numpy as np
import matplotlib.pyplot as plt
n = np.arange(-10, 11)
impulse = np.where(n == 0, 1, 0)
step = np.where(n >= 0, 1, 0)
ramp = np.where(n >= 0, n, 0)

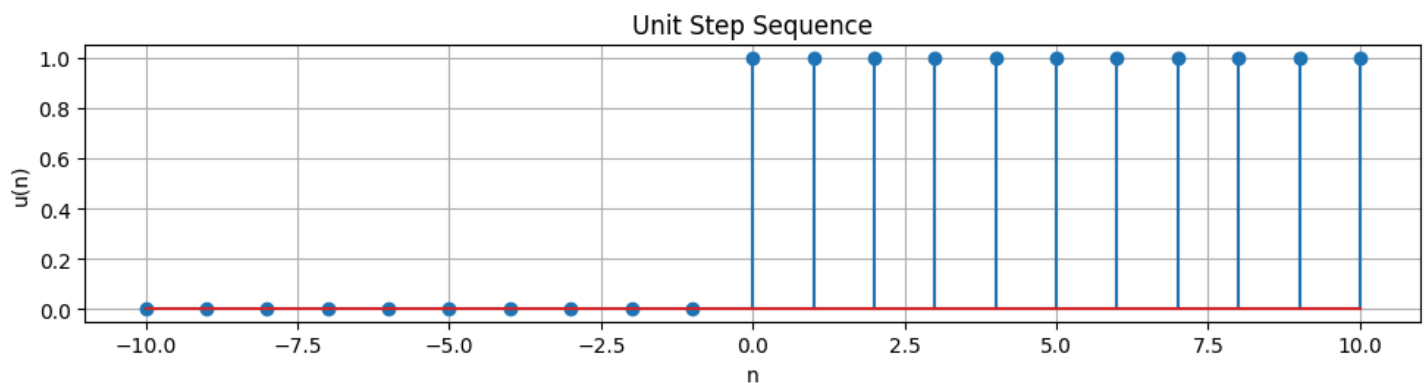
a = 0.8
exponential = np.where(n >= 0, a**n, 0)
f = 0.1 # Frequency
sinusoidal = np.sin(2 * np.pi * f * n)
plt.figure(figsize=(10, 12))

# Unit Step
plt.subplot(5, 1, 2)
plt.stem(n, step)
plt.title("Unit Step Sequence")
plt.xlabel("n")
plt.ylabel("u(n)")
plt.grid(True)

plt.tight_layout()
plt.show()

```

Output:



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Code 3:

```

import numpy as np
import matplotlib.pyplot as plt
n = np.arange(-10, 11)
impulse = np.where(n == 0, 1, 0)
step = np.where(n >= 0, 1, 0)
ramp = np.where(n >= 0, n, 0)

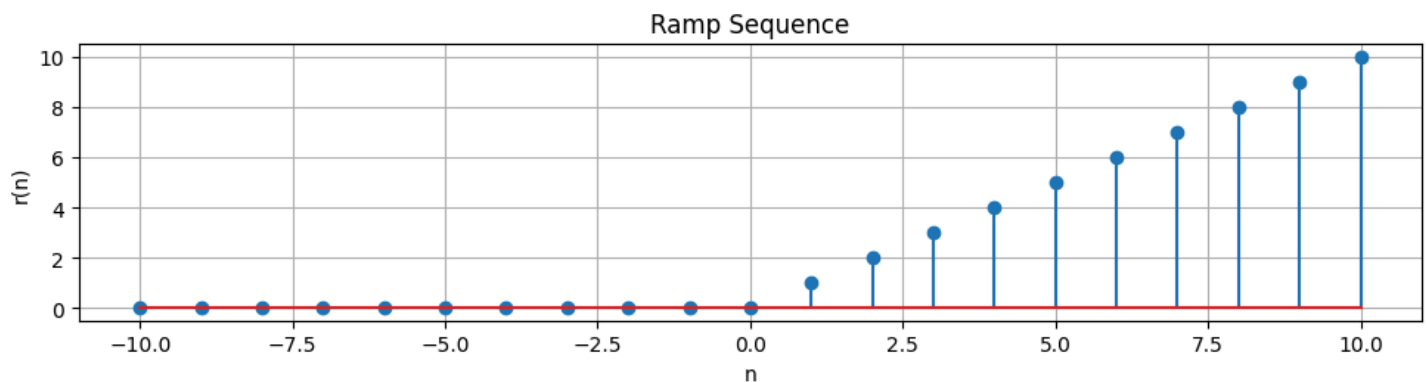
a = 0.8
exponential = np.where(n >= 0, a**n, 0)
f = 0.1 # Frequency
sinusoidal = np.sin(2 * np.pi * f * n)
plt.figure(figsize=(10, 12))

# Ramp
plt.subplot(5, 1, 3)
plt.stem(n, ramp)
plt.title("Ramp Sequence")
plt.xlabel("n")
plt.ylabel("r(n)")
plt.grid(True)

plt.tight_layout()
plt.show()

```

Output:



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Code 4:

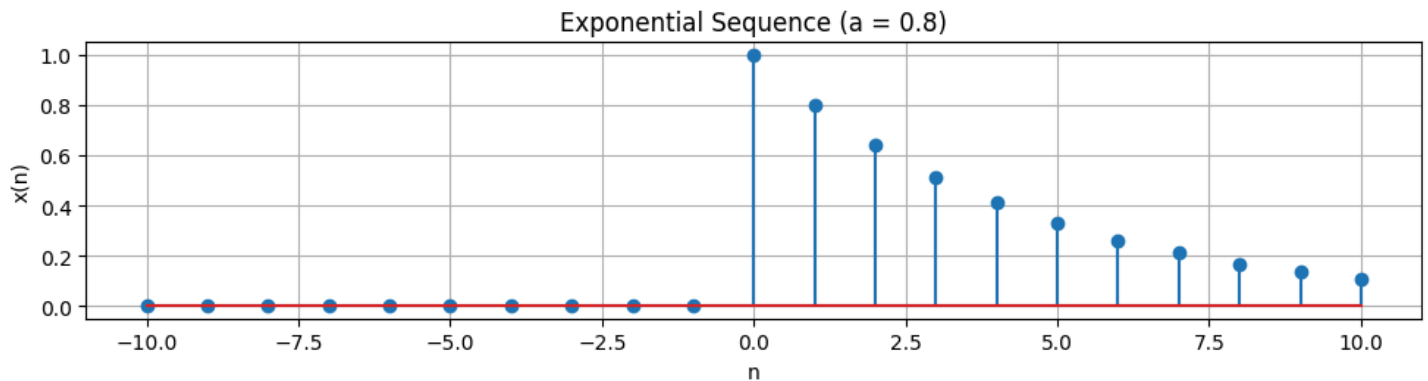
```
import numpy as np
import matplotlib.pyplot as plt
n = np.arange(-10, 11)
impulse = np.where(n == 0, 1, 0)
step = np.where(n >= 0, 1, 0)
ramp = np.where(n >= 0, n, 0)

a = 0.8
exponential = np.where(n >= 0, a**n, 0)
f = 0.1 # Frequency
sinusoidal = np.sin(2 * np.pi * f * n)
plt.figure(figsize=(10, 12))

# Exponential
plt.subplot(5, 1, 4)
plt.stem(n, exponential)
plt.title("Exponential Sequence (a = 0.8)")
plt.xlabel("n")
plt.ylabel("x(n)")
plt.grid(True)

plt.tight_layout()
plt.show()
```

Output:



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Code 5:

```
import numpy as np
import matplotlib.pyplot as plt
n = np.arange(-10, 11)
impulse = np.where(n == 0, 1, 0)
step = np.where(n >= 0, 1, 0)
ramp = np.where(n >= 0, n, 0)

a = 0.8
exponential = np.where(n >= 0, a**n, 0)
f = 0.1 # Frequency
sinusoidal = np.sin(2 * np.pi * f * n)
plt.figure(figsize=(10, 12))

# Sinusoidal
plt.subplot(5, 1, 5)
plt.stem(n, sinusoidal)
plt.title("Discrete-Time Sinusoidal Sequence")
plt.xlabel("n")
plt.ylabel("x(n)")
plt.grid(True)

plt.tight_layout()
plt.show()
```

Output:

